A new data structure for monomial ideals

Théo Ternier

Joint work with Pierre Lairez & Rafael Mohr

MATHEXP, Inria, France

ODELIX meeting

24th November 2025

Plan of the talk

Motivation: Signature Gröbner basis computations

Maximal sharing

Application 1: Hilbert series

Application 2: In commutative algebra

Processing of signatures

eco^{14}: 14 polynomials in 14 variables

(F5 algorithm)

eco^{14}: 14 polynomials in 14 variables

(F5 algorithm)

Number of membership tests in a monomial ideal:

117,861,802

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle, \qquad m = x^2y^2z^2 \in I?$

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle, \qquad m = x^2y^2z^2 \in I?$

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle, \qquad m = x^2y^2z^2 \in I?$

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle, \qquad m = x^2y^2z^2 \in I?$

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle, \qquad m = x^2y^2z^2 \in I?$

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle, \qquad m = x^2y^2z^2 \in I?$

Complexity naive algorithm

If n number of variables and N number of generators of I then:

$$O(n\,N)$$

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle, \qquad m = x^2y^2z^2 \in I?$

Complexity naive algorithm

If n number of variables and N number of generators of I then:

$$O(n\,N)$$

Complexity with our data structure

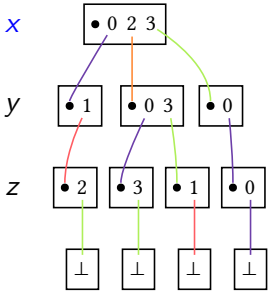
If n number of variables and N number of generators of I then:

$$O(n\,\ln(N))$$

State of the art

- Milowski in 2004: Prefix tree.

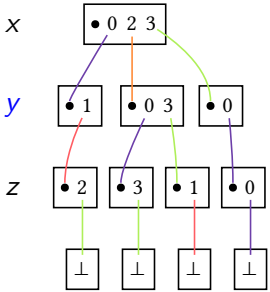
$I = \langle x^0yz^2, x^2z^3, x^2y^3z, x^3 \rangle \quad m = x^2y^2z^2 \in I$?



State of the art

- Milowski in 2004: Prefix tree.

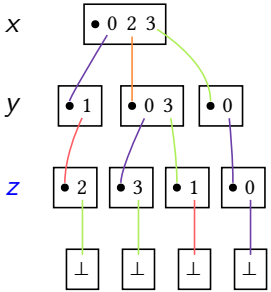
$I = \langle y^1 z^2, x^2 y^0 z^3, x^2 y^3 z, x^3 y^0 \rangle \quad m = x^2 y^2 z^2 \in I$?



State of the art

- Milowski in 2004: Prefix tree.

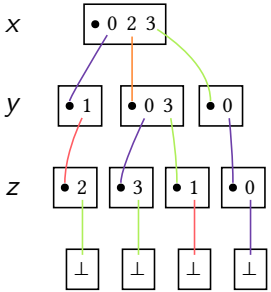
$I = \langle yz^2, x^2z^3, x^2y^3z, x^3z^0 \rangle \quad m = x^2y^2z^2 \in I$?



State of the art

- Milowski in 2004: Prefix tree.

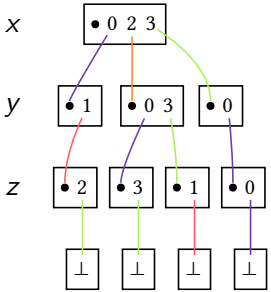
$I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle \quad m = x^2y^2z^2 \in I$?



State of the art

- Milowski in 2004: Prefix tree.

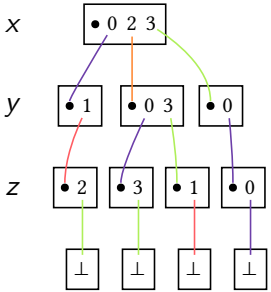
$I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle \quad m = x^2y^2z^2 \in I$?



State of the art

- Milowski in 2004: Prefix tree.

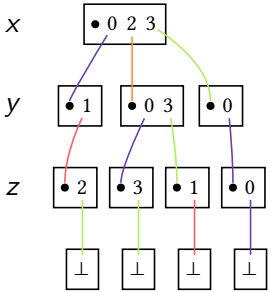
$I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle \quad m = x^2y^2z^2 \in I$?



State of the art

- Milowski in 2004: Prefix tree.

$I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle \quad m = x^2y^2z^2 \in I$?



- Roune and Stillman in 2012: kd-trees.

Notation

- ${x_1}^{\alpha_1} \dots {x_n}^{\alpha_n} \quad \rightsquigarrow \quad (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$

Notation

- ${x_1}^{\alpha_1} \dots {x_n}^{\alpha_n}$ $\rightsquigarrow$ $(\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$
- I a monomial ideal $\rightsquigarrow$ a subset of $\mathbb{N}^n$ such that $I = I + \mathbb{N}^n$

Notation

- ${x_1}^{\alpha_1} \dots {x_n}^{\alpha_n}$ $\rightsquigarrow$ $(\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$
- I a monomial ideal $\rightsquigarrow$ a subset of $\mathbb{N}^n$ such that $I = I + \mathbb{N}^n$
- Let $I \subset \mathbb{N}^n$ and $e \geq 0$ $\rightsquigarrow$ $I/e := \left\{\alpha \in \mathbb{N}^{n-1} \mid e \cdot \alpha \in I\right\}$

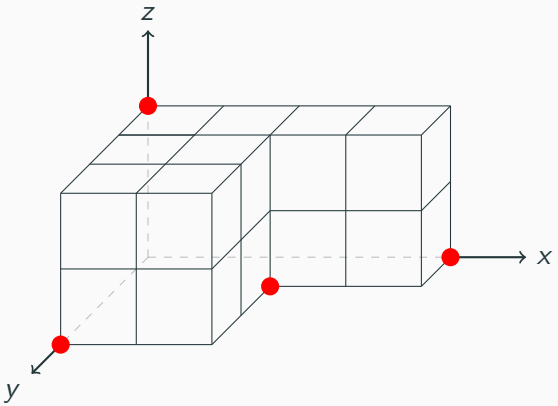
Notation

- ${x_1}^{\alpha_1} \dots {x_n}^{\alpha_n}$ $\rightsquigarrow$ $(\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$
- I a monomial ideal $\rightsquigarrow$ a subset of $\mathbb{N}^n$ such that $I = I + \mathbb{N}^n$
- Let $I \subset \mathbb{N}^n$ and $e \geq 0$ $\rightsquigarrow$ $I/e := \left\{\alpha \in \mathbb{N}^{n-1} \mid e \cdot \alpha \in I\right\}$

Equivalently: $I/e = (I : {x_1}^e) \cap \mathbb{Q}[x_2, \dots, x_n]$

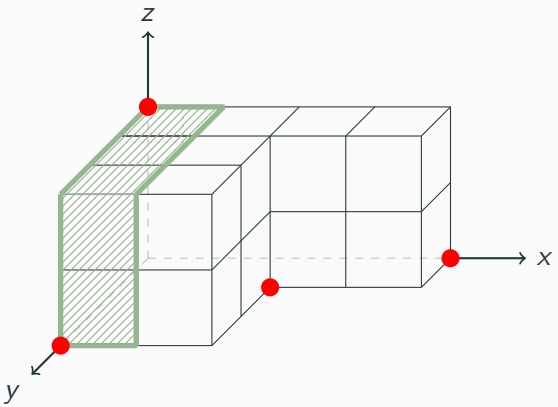
Staircase of a monomial ideal

Staircase of $I = \langle (0,3,0), (0,0,2), (2,1,0), (4,0,0) \rangle$



Staircase of a monomial ideal

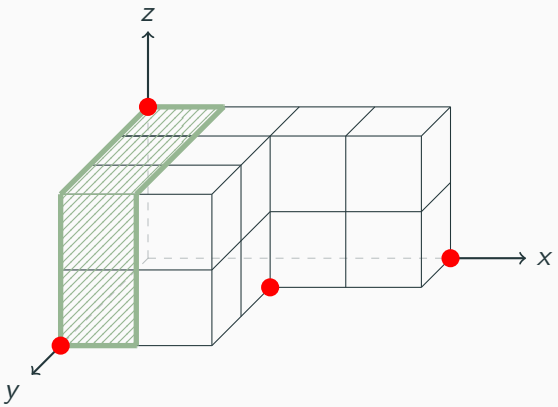
Staircase of $I = \langle (0,3,0),(0,0,2),(2,1,0),(4,0,0) \rangle$



$I/0 = \langle (3,0),(0,2) \rangle$

Staircase of a monomial ideal

Staircase of $I = \langle (0,3,0), (0,0,2), (2,1,0), (4,0,0) \rangle$



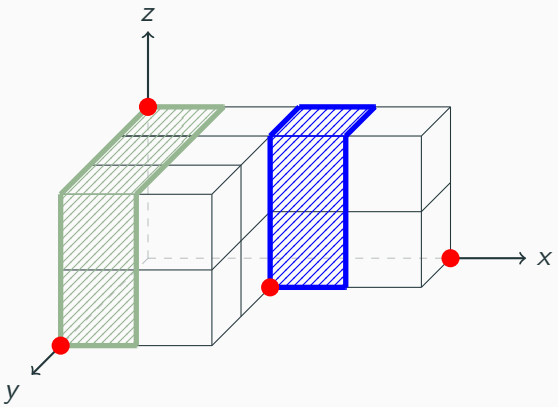
$$I/0 = \langle (3,0), (0,2) \rangle$$

$$\rotatebox{90}{$\subseteq$}$$

$$I/1 = \langle (3,0), (0,2) \rangle$$

Staircase of a monomial ideal

Staircase of $I = \langle (0,3,0),(0,0,2),(2,1,0),(4,0,0) \rangle$



$$I/0 = \langle (3,0),(0,2) \rangle$$

$$\rotatebox{90}{$\subseteq$}$$

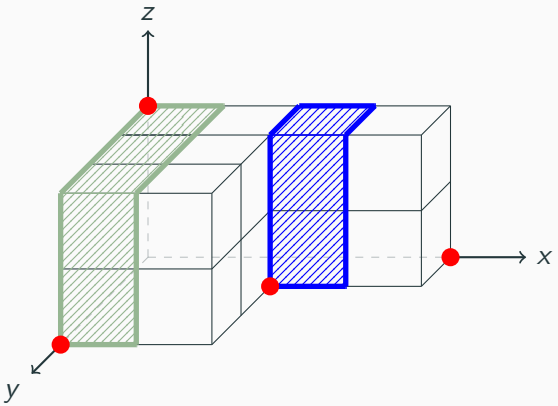
$$I/1 = \langle (3,0),(0,2) \rangle$$

$$\rotatebox{90}{$\subseteq$}$$

$$I/2 = \langle (3,0),(0,2),(1,0) \rangle$$

Staircase of a monomial ideal

Staircase of $I = \langle (0,3,0),(0,0,2),(2,1,0),(4,0,0) \rangle$



$$I/0 = \langle (3,0),(0,2) \rangle$$

$$\rotatebox{90}{$\subseteq$}$$

$$I/1 = \langle (3,0),(0,2) \rangle$$

$$\rotatebox{90}{$\subseteq$}$$

$$I/2 = \langle (3,0),(0,2),(1,0) \rangle$$

$$\rotatebox{90}{$\subseteq$}$$

$$I/3 = \langle (3,0),(0,2),(1,0) \rangle$$

Notion of n-trees

Let $\mathbb{N} \rightharpoonup \mathcal{X}$ the set of partial functions with finite domain.

Notion of n-trees

Let $\mathbb{N} \rightharpoonup \mathcal{X}$ the set of partial functions with finite domain.

n-trees

The set $\mathcal{T}_n$ of n-trees is defined recursively as:

Notion of n-trees

Let $\mathbb{N} \rightharpoonup \mathcal{X}$ the set of partial functions with finite domain.

n-trees

The set $\mathcal{T}_n$ of n-trees is defined recursively as:

- $\mathcal{T}_0 = \{\perp\}$

Notion of n-trees

Let $\mathbb{N} \rightharpoonup \mathcal{X}$ the set of partial functions with finite domain.

n-trees

The set $\mathcal{T}_n$ of n-trees is defined recursively as:

- $\mathcal{T}_0 = \{\perp\}$
- $\mathcal{T}_n = \{t : \mathbb{N} \rightharpoonup \mathcal{T}_{n-1} \mid \mathrm{dom}(t) \neq \emptyset\}$

Notion of n-trees

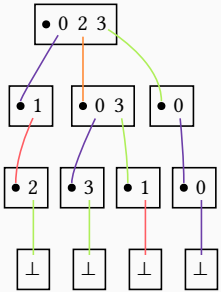
Let $\mathbb{N} \rightharpoonup \mathcal{X}$ the set of partial functions with finite domain.

n-trees

The set $\mathcal{T}_n$ of n-trees is defined recursively as:

- $\mathcal{T}_0 = \{\bot\}$
- $\mathcal{T}_n = \{t : \mathbb{N} \rightharpoonup \mathcal{T}_{n-1} \mid \mathrm{dom}(t) \neq \emptyset\}$

Example:



Notion of n-trees

Correspondance

There is a one-to-one correspondance between n-trees and non-empty finite subsets of $\mathbb{N}^n$:

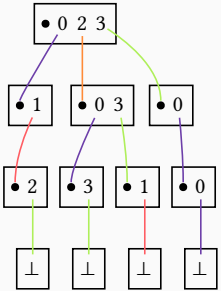
- $\mathrm{set}(\bot) = \mathbb{N}^0$
- $\mathrm{set}(t) = \bigcup_{e \geq 0} e \cdot \mathrm{set}\big(t(e)\big)$

Notion of n-trees

Correspondance

There is a one-to-one correspondance between n-trees and non-empty finite subsets of $\mathbb{N}^n$:

- $\mathrm{set}(\bot) = \mathbb{N}^0$
- $\mathrm{set}(t) = \bigcup_{e \geq 0} e \cdot \mathrm{set}\big(t(e)\big)$



$$\mathrm{set}(t) = \big\{(0,1,2),(2,0,3),(2,3,1),(3,0,0)\big\}$$

Monomial ideal represented by a n-tree

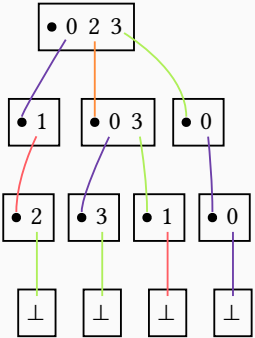
Ideal representation

A tree $t \in \mathcal{T}_n$ represents the monomial ideal $\mathbb{I}(t) = \langle \mathrm{set}(t) \rangle$

Monomial ideal represented by a n-tree

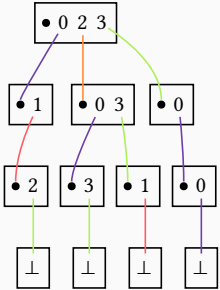
Ideal representation

A tree $t \in \mathcal{T}_n$ represents the monomial ideal $\mathbb{I}(t) = \langle \mathrm{set}(t) \rangle$

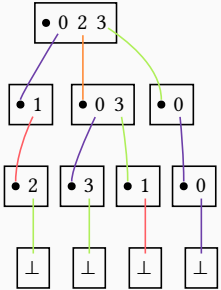


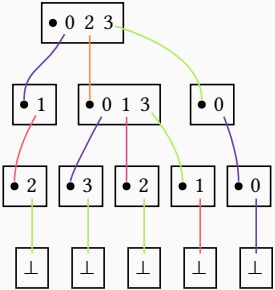
$$\mathbb{I}(t) = \langle (0,1,2), (2,0,3), (2,3,1), (3,0,0) \rangle$$

Ideal n-trees



Ideal n-trees



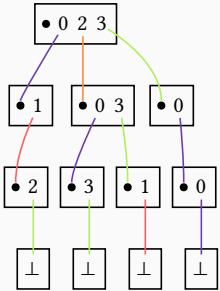


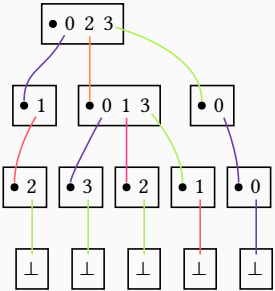
Ideal n-trees

Ideal n-trees

A tree $t \in \mathcal{T}_n$ is ideal whether $n = 0$ or $n > 0$ and

- $\forall e \in \mathrm{dom}(t), \qquad t(e)$ is ideal.
- $\forall e, f \in \mathrm{dom}(t), \qquad e < f \quad \Longrightarrow \quad \mathbb{I}(t(e)) \subsetneq \mathbb{I}(t(f))$





Membership in an ideal n-tree

def `contains`(t, α)

Input: An ideal n-tree t and $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$

Output: True if $\alpha \in \mathbb{I}(t)$ and false otherwise

```
if n = 0
    return True
s ← t(f) the maximal f ∈ dom(t) s.t. f ≤ α_1
if s = ∅
    return False
return contains(s, (α_2, ..., α_n))
```

Existence and uniqueness of ideal n-trees

Correspondance

There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals in $\mathbb{N}^n$.

Existence and uniqueness of ideal n-trees

Correspondance

There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals in $\mathbb{N}^n$.

Proof:

$$I/0 \subseteq I/1 \subseteq I/2 \subseteq \dots$$

Existence and uniqueness of ideal n-trees

Correspondance

There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals in $\mathbb{N}^n$.

Proof:

$$I/0 \subseteq I/1 \subseteq I/2 \subseteq \dots$$

Take $(f_i)_{i \in [\![1,r]\!]}$ such that:

- $I/e = \emptyset$ for all $e < f_1$

Existence and uniqueness of ideal n-trees

Correspondance

There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals in $\mathbb{N}^n$.

Proof:

$$I/0 \subseteq I/1 \subseteq I/2 \subseteq \dots$$

Take $(f_i)_{i \in [\![1,r]\!]}$ such that:

- $I/e = \emptyset$ for all $e < f_1$
- $I/e = I/f_i$ for all $f_i \leq e < f_{i+1}$

Existence and uniqueness of ideal n-trees

Correspondance

There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals in $\mathbb{N}^n$.

Proof:

$$I/0 \subseteq I/1 \subseteq I/2 \subseteq \dots$$

Take $(f_i)_{i \in [\![1,r]\!]}$ such that:

- $I/e = \emptyset$ for all $e < f_1$
- $I/e = I/f_i$ for all $f_i \leq e < f_{i+1}$
- $I/e = I/f_r$ for all $e \geq f_r$

Existence and uniqueness of ideal n-trees

Correspondance

There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals in $\mathbb{N}^n$.

Proof:

$$I/0 \subseteq I/1 \subseteq I/2 \subseteq \dots$$

Take $(f_i)_{i \in [\![1,r]\!]}$ such that:

- $I/e = \emptyset$ for all $e < f_1$
- $I/e = I/f_i$ for all $f_i \leq e < f_{i+1}$
- $I/e = I/f_r$ for all $e \geq f_r$
- $I/f_i \neq I/f_{i+1}$

Existence and uniqueness of ideal n-trees

Correspondance

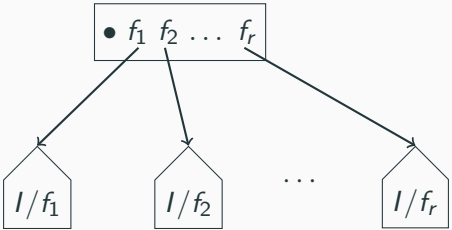
There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals in $\mathbb{N}^n$.

Proof:

$$I/0 \subseteq I/1 \subseteq I/2 \subseteq \dots$$

Take $(f_i)_{i \in [\![1,r]\!]}$ such that:

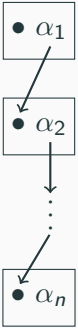
- $I/e = \emptyset$ for all $e < f_1$
- $I/e = I/f_i$ for all $f_i \leq e < f_{i+1}$
- $I/e = I/f_r$ for all $e \geq f_r$
- $I/f_i \neq I/f_{i+1}$



Insertion in an ideal n-tree

Let $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$.

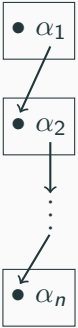
- Let `singleton`(α) be the ideal n-tree representing $\langle \alpha \rangle$:



Insertion in an ideal n-tree

Let $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$.

- Let $\texttt{singleton}(\alpha)$ be the ideal n-tree representing $\langle \alpha \rangle$:



- Let $\texttt{insert}(t, \alpha)$ be the ideal n-tree representing $\mathbb{I}(t) + \langle \alpha \rangle$.

Insertion in an ideal n-tree

Two important observations if $\alpha = x \cdot \alpha'$:

- For all $e < x$,

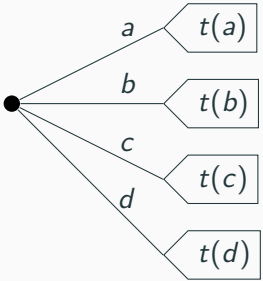
$$(\mathbb{I}(t) + \langle \alpha \rangle)/e \quad = \quad \mathbb{I}(t)/e$$

- For all $e \geq x$,

$$(\mathbb{I}(t) + \langle \alpha \rangle)/e \quad = \quad \mathbb{I}(t)/e + \langle \alpha' \rangle$$

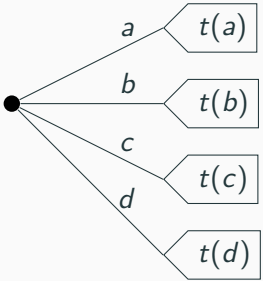
Insertion of $\alpha = x \cdot \alpha'$:

before insertion

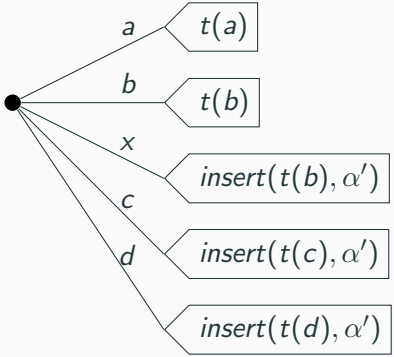


Insertion of $\alpha = x \cdot \alpha'$:

before insertion

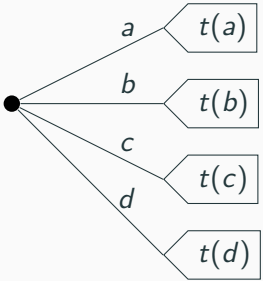


case $b < x < c$

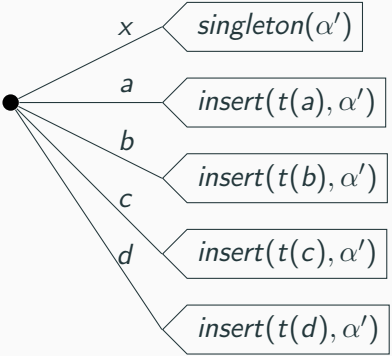


Insertion of $\alpha = x \cdot \alpha'$:

before insertion

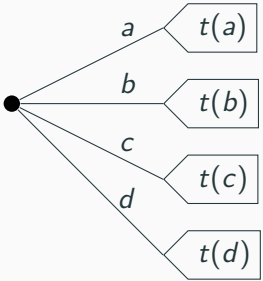


case $x < a$

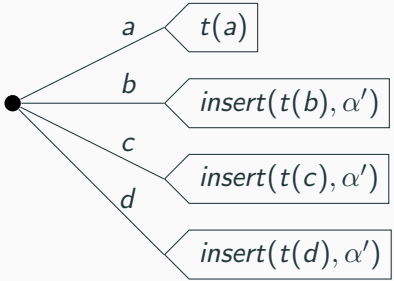


Insertion of $\alpha = x \cdot \alpha'$:

before insertion



case $x = b$



Insertion in an ideal n-tree

def `insert`(t, α)

1 **if** $t = \{\}$ **return** *singleton*(α)
2 **if** $n = 0$ **return** $\{\}$

Insertion in an ideal n-tree

def `insert`(t, α)

1 **if** $t = \{\}$ **return** *singleton*(α)
2 **if** $n = 0$ **return** $\{\}$

Insertion in an ideal n-tree

def `insert`(t, α)

1 **if** $t = \{\}$ **return** *singleton*(α)
2 **if** $n = 0$ **return** $\{\}$
3 $s \leftarrow \{\}$
4 Get x and α' s.t. $\alpha = x \cdot \alpha'$

Insertion in an ideal n-tree

def $\texttt{insert}(t, \alpha)$

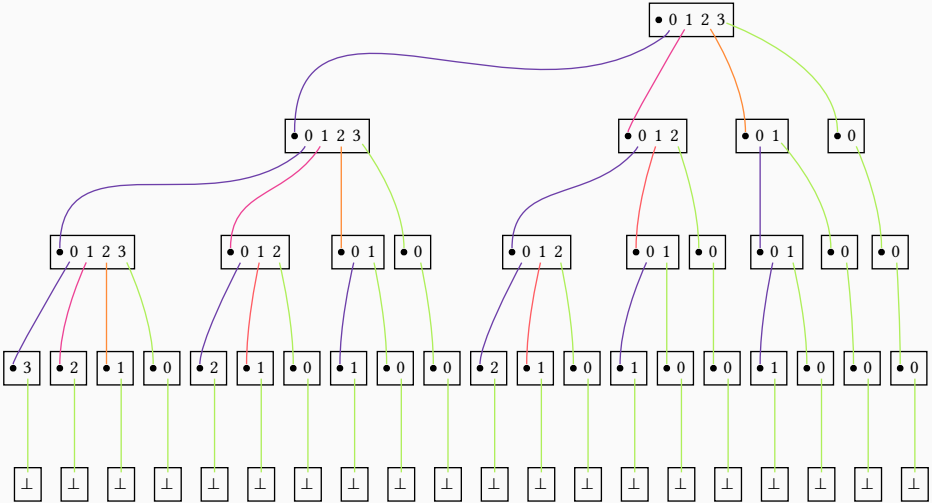
```
if t = {} return singleton(α)
if n = 0 return {}
s ← {}
Get x and α′ s.t. α = x · α′
for e ∈ dom(t)
        if e < x
                s(e) ← t(e)
        elif e > x
                s(e) ← insert(t(e), α′)
```

Insertion in an ideal n-tree

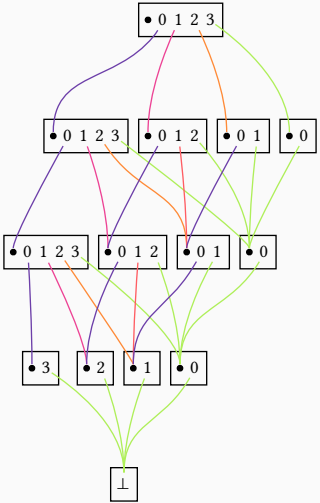
def $\texttt{insert}(t, \alpha)$

```
if t = {} return singleton(α)
if n = 0 return {}
s ← {}
Get x and α′ s.t. α = x · α′
for e ∈ dom(t)
        if e < x
                s(e) ← t(e)
        elif e > x
                s(e) ← insert(t(e), α′)
s(x) ← insert(t(f), α′) the maximal f ∈ dom(t) s.t. f ≤ x
remove f ∈ dom(s) s.t. s(f) = s(e) when e < f
return s
```

Problem of width for ideal n-trees



Solved problem of width with maximal sharing



Maximal Sharing

Invariant: Two trees are *structurally equal* iff they are *physically equal*.

Maximal Sharing

Invariant: Two trees are *structurally equal* iff they are *physically equal*.

- Identical subtrees are stored only once.

Maximal Sharing

Invariant: Two trees are *structurally equal* iff they are *physically equal*.

- Identical subtrees are stored only once.
- Memory savings by avoiding duplication.

Maximal Sharing

Invariant: Two trees are *structurally equal* iff they are *physically equal*.

- Identical subtrees are stored only once.
- Memory savings by avoiding duplication.
- Speed-up: equality checks are $\mathcal{O}(1) \implies$ memoization.

Maximal Sharing

Invariant: Two trees are *structurally equal* iff they are *physically equal*.

- Identical subtrees are stored only once.
- Memory savings by avoiding duplication.
- Speed-up: equality checks are $\mathcal{O}(1) \implies$ memoization.

Result: Manipulation of ideal n-trees much faster and with less memory.

One example (from Gröbner basis computations)

Number of gen: 5588 Number of nodes: 222

Signature Gröbner basis computations

- Buchberger in 1965: many reductions (S-polynomials)

Signature Gröbner basis computations

- Buchberger in 1965: many reductions (S-polynomials)
- Faugère in 2002: avoid useless operations with signatures

Signature Gröbner basis computations

- Buchberger in 1965: many reductions (S-polynomials)
- Faugère in 2002: avoid useless operations with signatures
- Observations:
  - Fewer arithmetic operations

Signature Gröbner basis computations

- Buchberger in 1965: many reductions (S-polynomials)
- Faugère in 2002: avoid useless operations with signatures
- Observations:
  - Fewer arithmetic operations
  - More symbolic preprocessing (membership tests)

Signature Gröbner basis computations

	AlgebraicSolving.jl	*AlgebraicSolving.jl* with mdd					
name	time	time	test	mdd	size	nbr test	$\times$
katsura^{11}	3.5s	1.8s	0.04s	0.02s	65	387705	1.9
katsura^{12}	26s	14s	0.16s	0.1s	81	1503655	1.9
katsura^{13}	3min	1min52	0.7s	0.5s	103	5863729	1.6
cyclic^{7}	0.95s	0.36s	0.02s	0.01s	107	239572	1.6
cyclic^{8}	1min	29s	0.5s	0.07s	350	4145249	2.1
eco^{14}	3h30	1h06	6s	32s	172	117861802	3.2

AlgebraicSolving.jl: Christian Eder (Kaiserslautern), Mohab Safey El Din (Sorbonne Université) and Rafael Mohr (Leuven).

Signature Gröbner basis computations

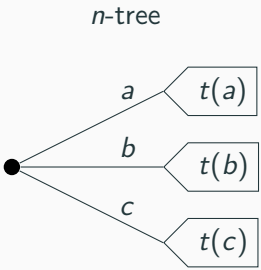
	AlgebraicSolving.jl	*AlgebraicSolving.jl* with mdd					
name	time	time	test	mdd	size	nbr test	×
katsura^{11}	3.5s	1.8s	0.04s	0.02s	65	387705	1.9
katsura^{12}	26s	14s	0.16s	0.1s	81	1503655	1.9
katsura^{13}	3min	1min52	0.7s	0.5s	103	5863729	1.6
cyclic^{7}	0.95s	0.36s	0.02s	0.01s	107	239572	1.6
cyclic^{8}	1min	29s	0.5s	0.07s	350	4145249	2.1
eco^{14}	3h30	1h06	6s	32s	172	117861802	3.2

AlgebraicSolving.jl: Christian Eder (Kaiserslautern), Mohab Safey El Din (Sorbonne Université) and Rafael Mohr (Leuven).

Another way to create ideal n-trees: `sat`

Input: A n-tree

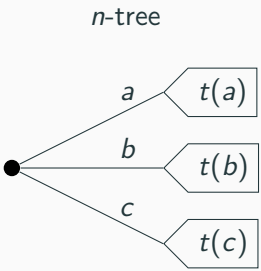
Output: The ideal n-tree representing $\mathbb{I}(t)$

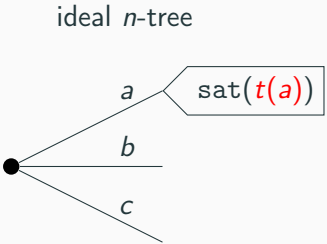


Another way to create ideal n-trees: `sat`

Input: A n-tree

Output: The ideal n-tree representing $\mathbb{I}(t)$

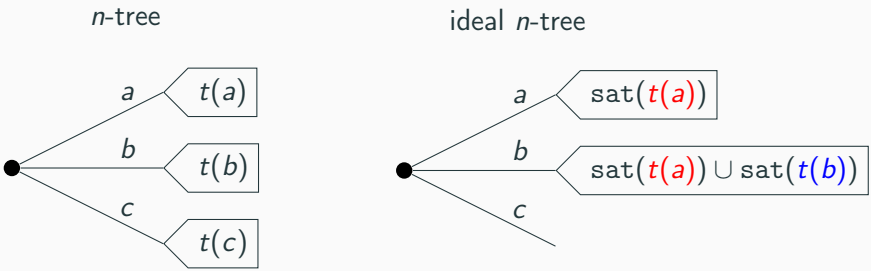




Another way to create ideal n-trees: `sat`

Input: A n-tree

Output: The ideal n-tree representing $\mathbb{I}(t)$

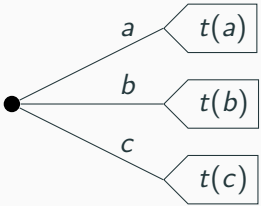


Another way to create ideal n-trees: `sat`

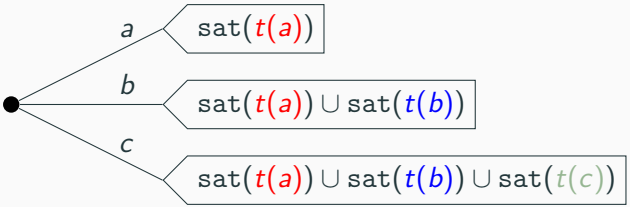
Input: A n-tree

Output: The ideal n-tree representing $\mathbb{I}(t)$

n-tree



ideal n-tree

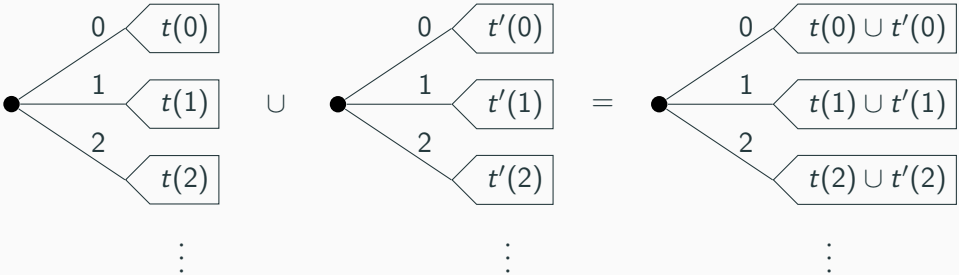


`merge` function

$$(I \cup J)/\alpha \quad = \quad I/\alpha \quad \cup \quad J/\alpha$$

`merge` function

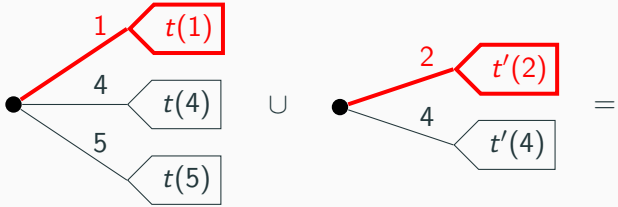
$$(I \cup J)/\alpha \quad = \quad I/\alpha \quad \cup \quad J/\alpha$$



`merge` function

$i = 0$ current index for T

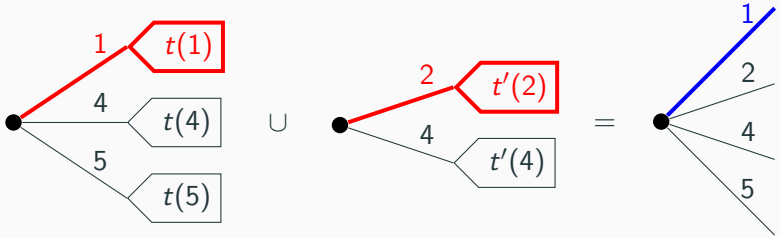
$j = 0$ current index for T'



`merge` function

$i = 0$ current index for T

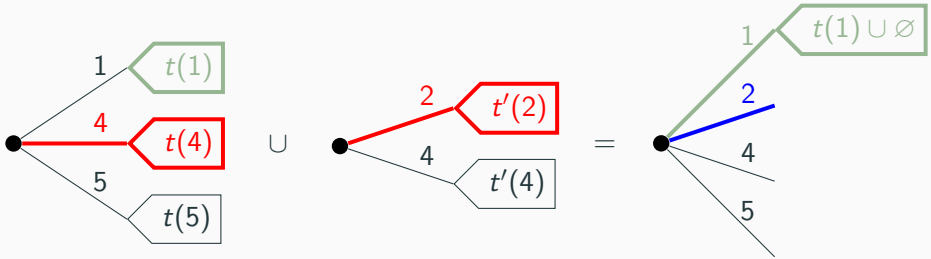
$j = 0$ current index for T'



`merge` function

$i = 1$ current index for T **(+1)**

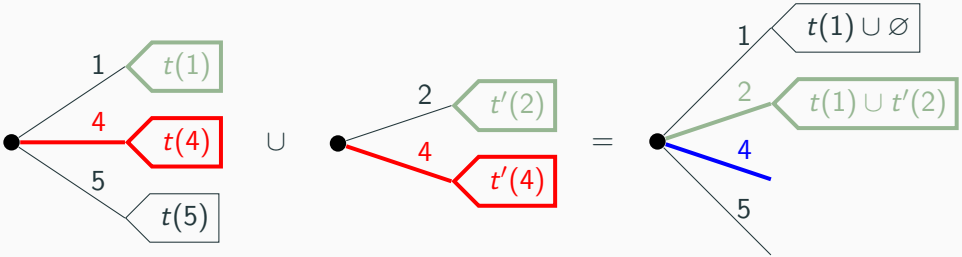
$j = 0$ current index for T'



`merge` function

$i = 1$ current index for T

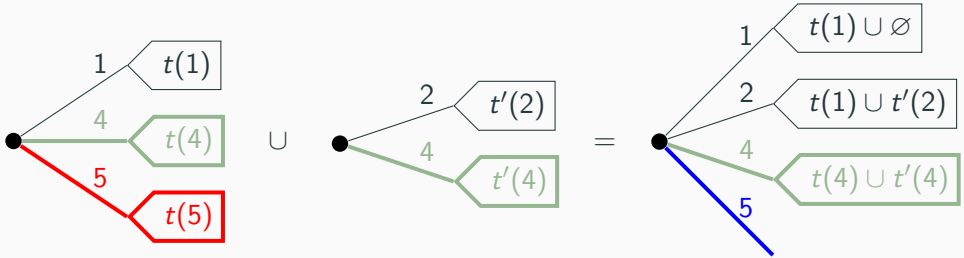
$j = 1$ current index for T' **(+1)**



`merge` function

$i = 2$ current index for T **(+1)**

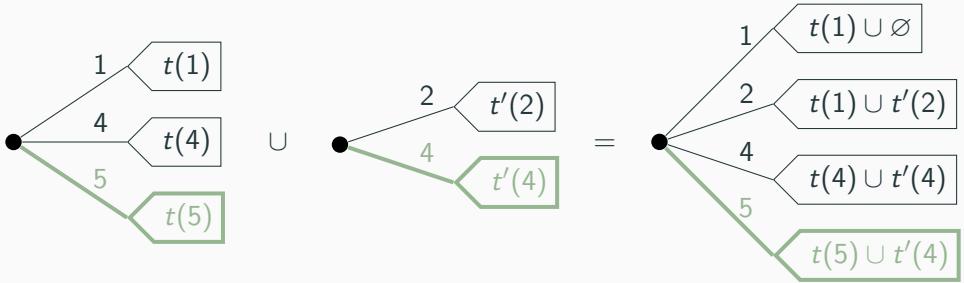
$j = 1$ current index for T'



`merge` function

$i = 2$ current index for T

$j = 1$ current index for T'



Another way to create n-trees: `sat`

def `sat`(t)

Input: A n-tree t

Output: The ideal n-tree representing $\mathbb{I}(t)$

```
s ← {}
prev ← None
for e ∈ dom(t) by increasing order
    s(e) ← sat(t(e))
    s(e) ← merge(prev, s(e))
    prev ← s(e)
return s
```

Application 1: Hilbert series

Hilbert series

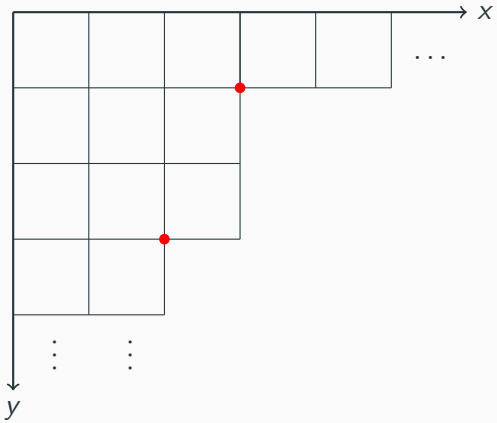
Let $d \in \mathbb{N}$, A_d the set of monomials of degree d outside of I.

The Hilbert series of I is:

$$H_I(t) := \sum_{d \geq 0} |A_d| \, t^d = \sum_{m \notin I} t^{\deg(m)}$$

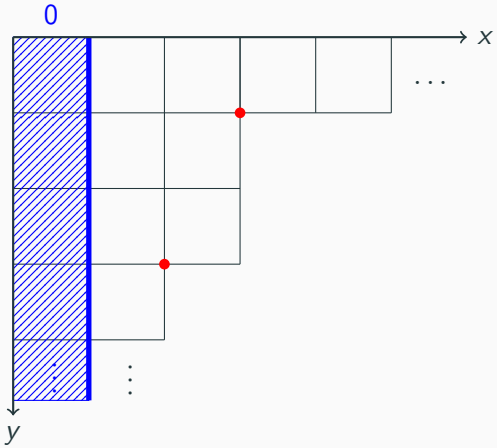
Notation: If m is the mdd of I then $H_m := H_I(t)$

Application 1: Hilbert series



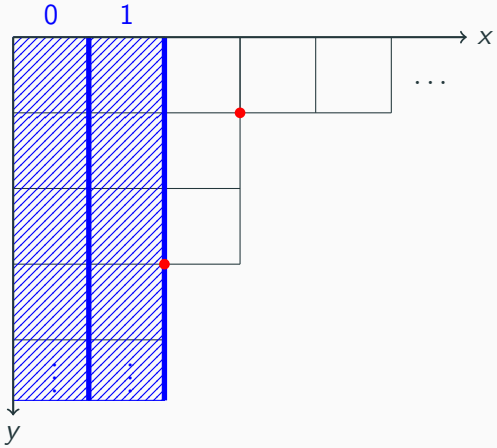
$$H_I(t) =$$

Application 1: Hilbert series



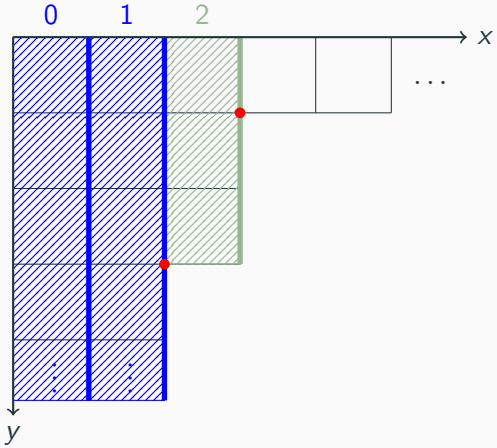
$$H_I(t) = t^0 \, H_\varnothing(t)$$

Application 1: Hilbert series



$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t)$$

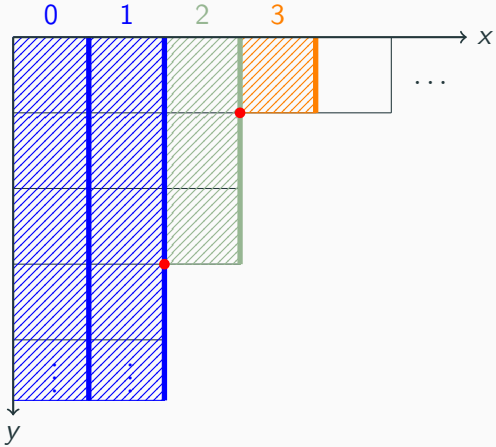
Application 1: Hilbert series



$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t)$$

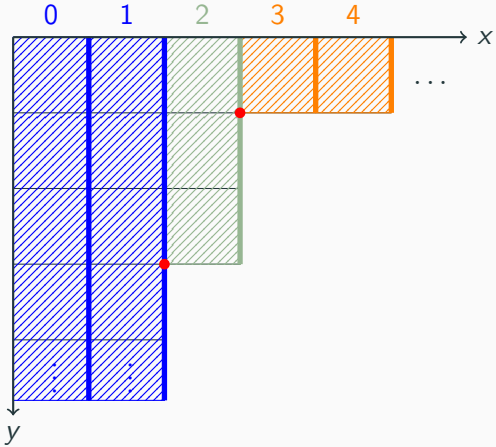
$$+\, t^2\, H_{\langle x^3 \rangle}(t)$$

Application 1: Hilbert series



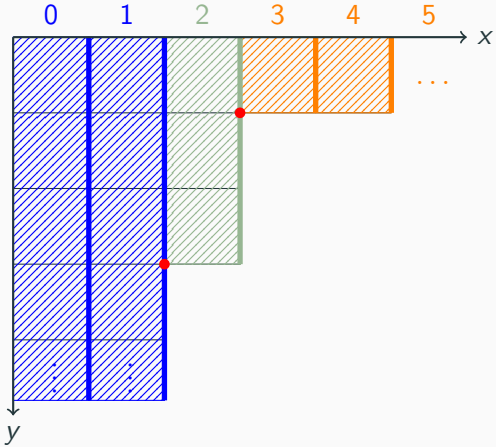
$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t)$$
$$+\, t^2\, H_{\langle x^3 \rangle}(t) + t^3\, H_{\langle x \rangle}(t)$$

Application 1: Hilbert series



$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t)$$
$$+ t^2\, H_{\langle x^3 \rangle}(t) + t^3\, H_{\langle x \rangle}(t)$$
$$+ t^4\, H_{\langle x \rangle}(t)$$

Application 1: Hilbert series



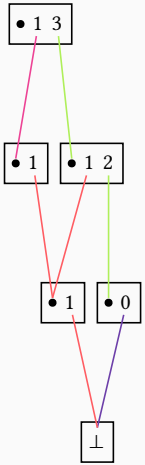
$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t)$$

$$+ t^2\, H_{\langle x^3 \rangle}(t) + t^3\, H_{\langle x \rangle}(t)$$

$$+ \sum_{k \geq 4} t^k\, H_{\langle x \rangle}(t)$$

Application 1: Hilbert series

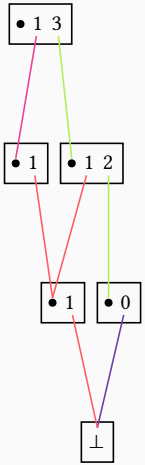
For $I = \langle xyz, x^3y^2 \rangle$:



Application 1: Hilbert series

For $I = \langle xyz, x^3y^2 \rangle$:

$$H_I(t) = t^0\, H_\varnothing + t^1\, H_{m(1)} + t^2\, H_{m(1)} + t^3\, H_{m(3)} + t^4\, H_{m(3)} + \dots$$

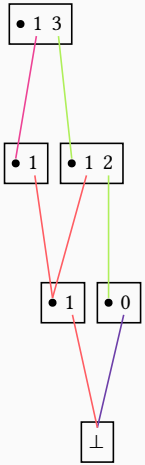


Application 1: Hilbert series

For $I = \langle xyz, x^3y^2 \rangle$:

$$H_I(t) = t^0\, H_\varnothing + t^1\, H_{m(1)} + t^2\, H_{m(1)} + t^3\, H_{m(3)} + t^4\, H_{m(3)} + \dots$$

$$= \sum_{k<1} t^k\, H_\varnothing + \sum_{1\le k<3} t^k\, H_{m(1)} + \sum_{k\ge 3} t^k\, H_{m(3)}$$



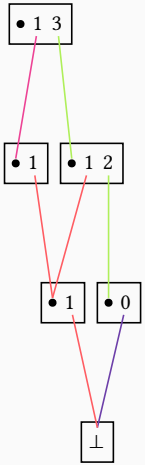
Application 1: Hilbert series

For $I = \langle xyz, x^3y^2 \rangle$:

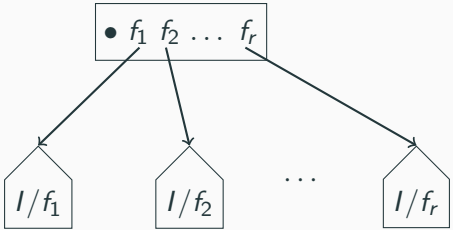
$$H_I(t) = t^0\, H_\varnothing + t^1\, H_{m(1)} + t^2\, H_{m(1)} + t^3\, H_{m(3)} + t^4\, H_{m(3)} + \dots$$

$$= \sum_{k<1} t^k\, H_\varnothing + \sum_{1\le k<3} t^k\, H_{m(1)} + \sum_{k\ge 3} t^k\, H_{m(3)}$$

$$= \frac{t^1}{(1-t)^2} + \frac{t^1 - t^3}{1-t}\, H_{m(1)} + \frac{t^3}{1-t}\, H_{m(3)}$$



Application 1: Hilbert series



$$H_I(t) = \frac{t^{f_1}}{(1-t)^n} + \sum_{k=1}^{r-1} \frac{t^{f_k} - t^{f_{k+1}}}{1-t} H_{m(f_k)} + \frac{t^{f_r}}{1-t} H_{m(f_r)}$$

Application 1: Hilbert series

			frobby	monid.rs		
name	gen	var	time	time	size	×
katsura[12]	1050	12	400ms	0.6ms	102	700
katsura[13]	2091	13	1s	1ms	124	1k
eco[14]	2852	14	4s	2ms	172	2k
V10g400	400	10	10s	1s	536329	10
V10g800	800	10	35s	3s	1106025	10
V10g1000	1000	10	50s	4s	1461567	10
frobn13d11	170835	13	18h	11min	34763212	100
J51	3036	89	>1d	14ms	1605	>6m
J60	3432	89	>1d	17ms	1798	>5m
K5	13313	32	>1d	10s	1098244	>8k
tcyc5d25p	3000	125	>1d	16ms	1181	>5m
tcyc5d30p	4350	150	>1d	25ms	1431	>3m
V20g100	100	20	12ms	5ms	6374	2

Application 1: Hilbert series

			frobby	monid.rs		
name	gen	var	time	time	size	×
katsura[12]	1050	12	400ms	0.6ms	102	700
katsura[13]	2091	13	1s	1ms	124	1k
eco[14]	2852	14	4s	2ms	172	2k
V10g400	400	10	10s	1s	536329	10
V10g800	800	10	35s	3s	1106025	10
V10g1000	1000	10	50s	4s	1461567	10
frobn13d11	170835	13	18h	11min	34763212	100
J51	3036	89	>1d	14ms	1605	>6m
J60	3432	89	>1d	17ms	1798	>5m
K5	13313	32	>1d	10s	1098244	>8k
tcyc5d25p	3000	125	>1d	16ms	1181	>5m
tcyc5d30p	4350	150	>1d	25ms	1431	>3m
V20g100	100	20	12ms	5ms	6374	2

Application 2: Duality

Idea: Write I a monomial ideal as intersection of irreducible ideals.

Application 2: Duality

Idea: Write I a monomial ideal as intersection of irreducible ideals.

Irreducible 0-dimensional monomial ideals

All of the form:

$$\mathfrak{m}_{\underline{a}} = \left\langle {x_1}^{a_1}, \ldots, {x_n}^{a_n} \right\rangle$$

where $a_i \in \mathbb{N}$

Application 2: Duality

Idea: Write I a monomial ideal as intersection of irreducible ideals.

Irreducible 0-dimensional monomial ideals

All of the form:

$$\mathfrak{m}_{\underline{a}} = \left\langle {x_1}^{a_1}, \ldots, {x_n}^{a_n} \right\rangle$$

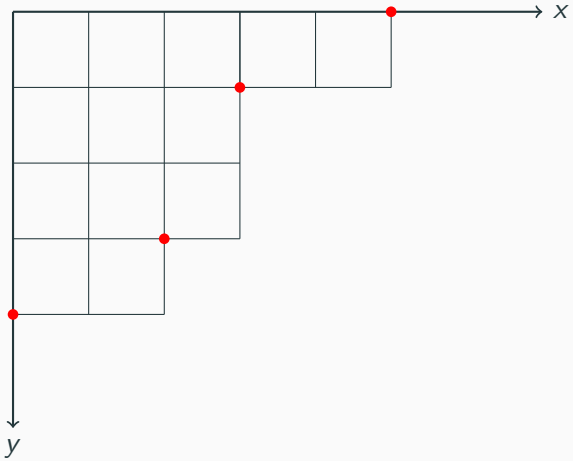
where $a_i \in \mathbb{N}$

Irreducible components

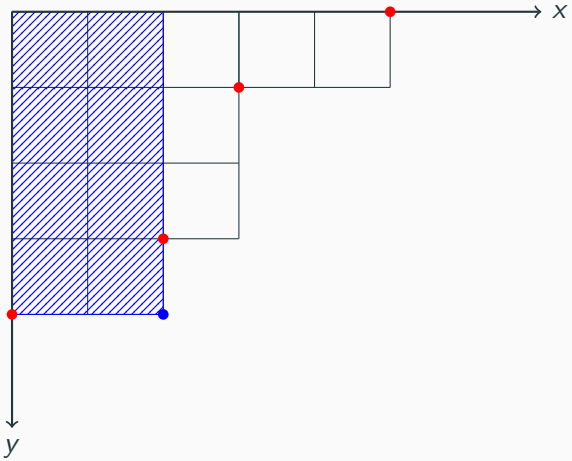
There exists unique $(\underline{a}_1, \ldots, \underline{a}_r)$ s.t.

$$I = \bigcap_{j=1}^{r} \mathfrak{m}_{\underline{a}_j}$$

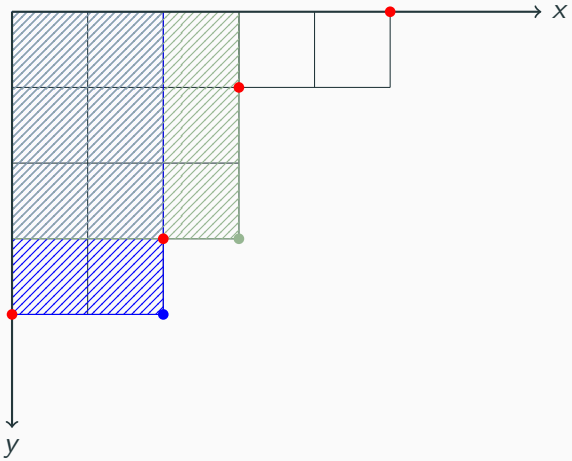
Application 2: Duality



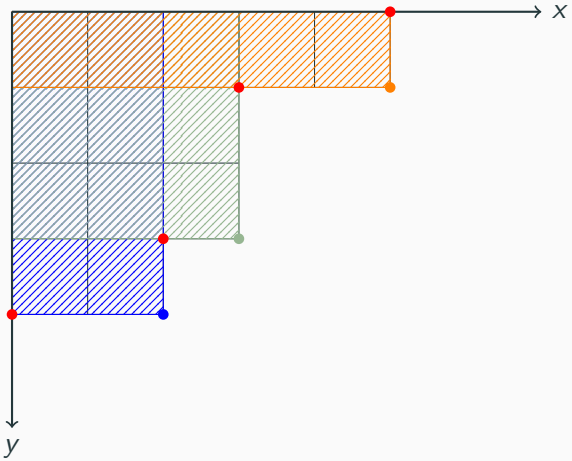
Application 2: Duality



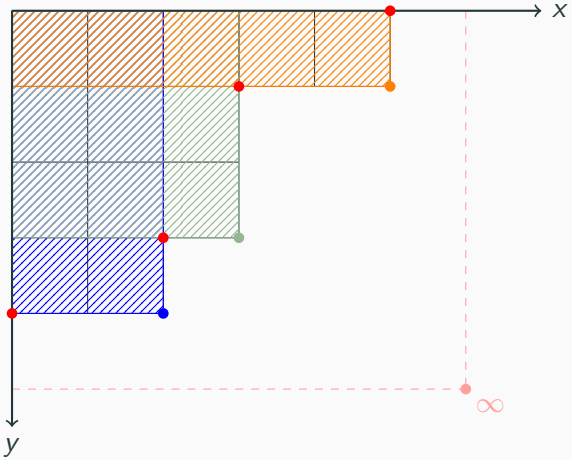
Application 2: Duality



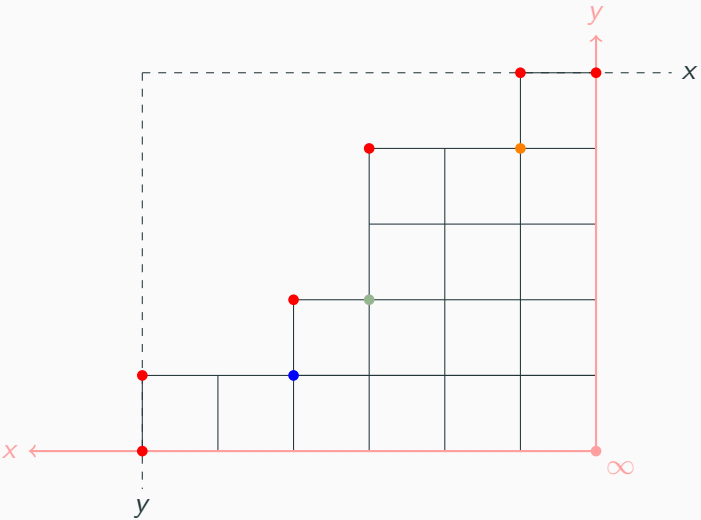
Application 2: Duality



Application 2: Duality

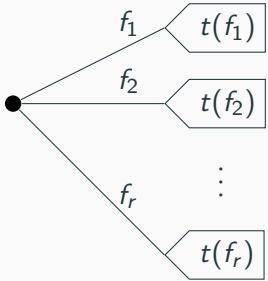


Application 2: Duality



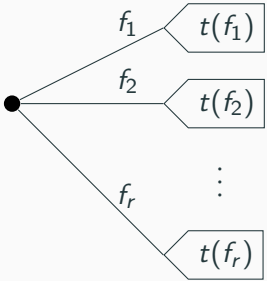
Application 2: Duality

ideal n-tree

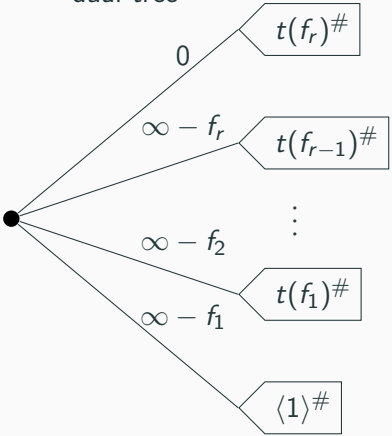


Application 2: Duality

ideal n-tree



dual tree



Perspectives and work in progress

- Get a bound on the size of mdds.
- Find the index of the divisor.
- Free resolutions.